

Lecture 9: Reservoir Computing
Echo State Networks, Spiking Neurons, Cellular Automata

Deep Learning in Classical and Quantum Physics

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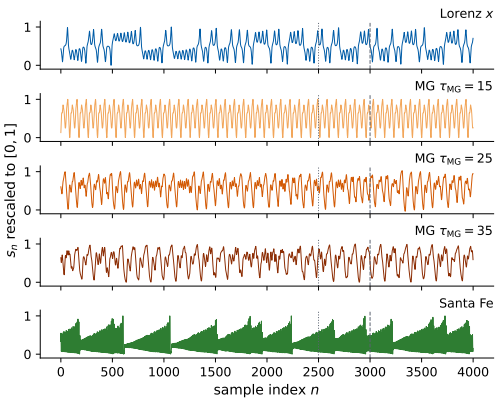
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Recap



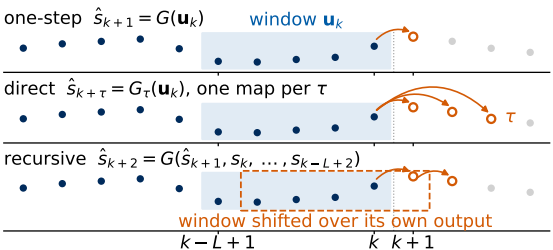
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Five Records



- Lorenz: x of the Lorenz-63 convection model, chaotic
- Mackey–Glass (MG) delay equation, delay $\tau_{\mathrm{MG}} = 15$ periodic, 25 and 35 chaotic
- Santa Fe: far-infrared laser, pulsations collapsing at irregular times
- 4000 samples in $[0, 1]$: training 2500, validation 500, test 1000 (dashed)
- choices on validation; test used once

Forecast Task



Forecast at horizon τ

$$\hat{s}_{k+\tau} = G_\tau(s_k, s_{k-1}, \ldots)$$

G_τ: model τ steps ahead; $\mathbf{u}_k$: window of past values; direct: one output per horizon; recursive: one-step model fed its own output

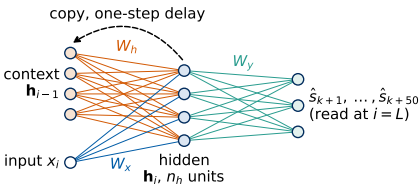
Capacity

$$C(\tau) = r^2(\hat{s}_{k+\tau}, s_{k+\tau})$$

$$C_{\text{tot}} = \sum_{\tau=1}^{50} C(\tau)$$

r: Pearson correlation over the test origins k; $C_{\text{tot}} \leq 50$; baseline: persistence $\hat{s}_{k+\tau} = s_k$

Recurrent Network with All Weights Trained



Recurrent neural network (RNN)

$$\mathbf{h}_k = \tanh\left(W_h \mathbf{h}_{k-1} + W_x s_k + \mathbf{b}\right), \quad \hat{\mathbf{y}}_k = W_y \mathbf{h}_k + \mathbf{b}_y$$

$\mathbf{h}_k \in \mathbb{R}^{n_h}$: hidden state of n_h units; $W_h \in \mathbb{R}^{n_h \times n_h}$: feedback; $W_x \in \mathbb{R}^{n_h}$: input; $W_y, \mathbf{b}_y$: readout of the 50 horizons

- the state $\mathbf{h}_k$ is fed back at the next step, so it depends on $s_k, s_{k-1}, \ldots$
- all weights trained through time: gradients vanish or explode, training is slow

J. L. Elman, Cogn. Sci. 14, 179 (1990), doi:10.1207/s15516709cog1402_1

Keep the Recurrence, Drop the Training

RNN $\mathbf{h}_k = \tanh\left(W_h\,\mathbf{h}_{k-1} + W_x\,s_k\right)$ $\qquad \hat{s}_{k+\tau} = W_y\,\mathbf{h}_k$

reservoir $\mathbf{r}_k = \tanh\left(W\,\mathbf{r}_{k-1} + \mathbf{w}_{\text{in}}\,s_k\right)$ $\qquad \hat{s}_{k+\tau} = \mathbf{w}_\tau\,\mathbf{r}_k$

trained random, fixed

- same recurrence: the state remembers the past input
- RNN: every weight learned by backpropagation through time, slow and unstable
- reservoir: only the readout learned, by one linear solve

Physics Knowledge in the Model

model	known	learned
AR, NVAR	only the record s_k	readout of fixed functions of past values
MLP, RNN, LSTM	only the record s_k	features and readout
Neural ODE	form $\dot{\mathbf{x}} = \mathbf{F}(\mathbf{x})$	vector field $\mathbf{F}_\theta$
PINN	equation, initial state	solution $x_\theta(t)$
reservoir	only the record s_k	linear readout of a fixed random state
cellular automaton	only the record s_k	linear readout of automaton states
hybrid model	part of $\mathbf{F}$	missing term $\mathbf{U}_\theta$, then a formula

- reservoir (bold): the features of an RNN, but random and never trained
- cellular automaton: the same idea on a discrete substrate

AR: autoregressive; NVAR: nonlinear vector autoregression; MLP: multilayer perceptron; LSTM: long short-term memory; ODE: ordinary differential equation; PINN: physics-informed neural network

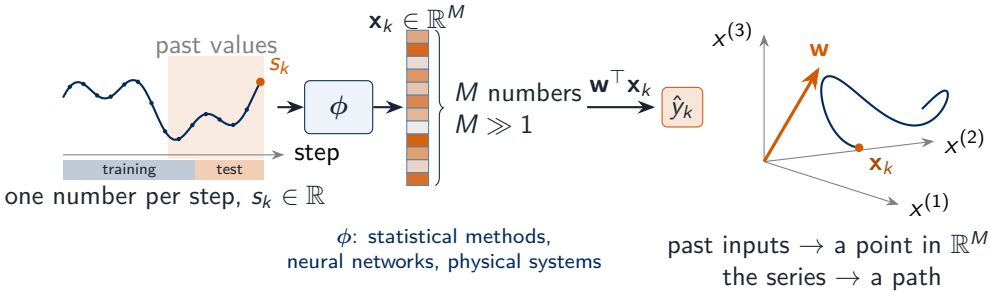
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Reservoir Computing

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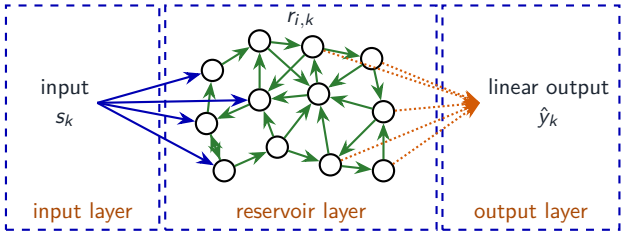
Time Series in a Feature Space, Linear Readout

$$(s_1, \ldots, s_k) \xrightarrow{\phi} \mathbf{x}_k \in \mathbb{R}^M \xrightarrow{\text{linear}} \hat{y}_k = \mathbf{w}^\top \mathbf{x}_k$$



- AR, NVAR: ϕ fixed, explicit function of the last values; RNN: state $\mathbf{x}_k = \mathcal{F}(\mathbf{x}_{k-1}, s_k)$, $\mathcal{F}$ trained
- reservoir: state $\mathbf{r}_k = \mathcal{F}(\mathbf{r}_{k-1}, s_k)$ with $\mathcal{F}$ fixed, the features $\mathbf{x}_k = \mathbf{r}_k$; only $\mathbf{w}$ trained

Echo State Network (ESN)



Idea

keep the nonlinear feature map fixed, train only the linear readout $\mathbf{w}$: one least-squares fit

- N nodes coupled by random recurrent weights W, driven through random input weights $\mathbf{w}_{\text{in}}$, both fixed
- the state $\mathbf{r}_k$: a nonlinear echo of the input history
- readout $\hat{y}_k = \mathbf{w}^\top \mathbf{r}_k$: the only trained part

H. Jaeger, GMD Report 148 (2001)

State Update and Readout

Leaky state update

$$\mathbf{r}_k = (1-\alpha)\,\mathbf{r}_{k-1} + \alpha \tanh\left(W\mathbf{r}_{k-1} + \mathbf{w}_{\text{in}} s_k\right)$$

$\mathbf{r}_k \in \mathbb{R}^N$: state; W: random, sparse, spectral radius ρ (largest |eigenvalue|); $\mathbf{w}_{\text{in}}$: uniform in $[-\sigma_{\text{in}}, \sigma_{\text{in}}]$; $\alpha \in (0, 1]$: leak rate

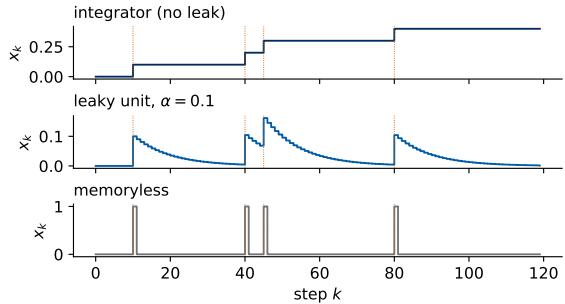
Readout by ridge regression

$$\min_{\mathbf{w}} \sum_k \left(y_k - \mathbf{w}^\top \mathbf{r}_k\right)^2 + \beta\|\mathbf{w}\|^2 \;\Rightarrow\; \mathbf{w} = (X^\top X + \beta \not{1})^{-1} X^\top \boldsymbol{y}$$

target $y_k = s_{k+\tau}$, collected in $\boldsymbol{y}$; rows of X: states $\mathbf{r}_k^\top$; washout: first states, which remember the start, left out; $\beta > 0$: ridge penalty; no gradient descent

- α, ρ: how long the state remembers ($\alpha = 1$: plain tanh); $N, \rho, \alpha, \sigma_{\text{in}}, \beta$ chosen on the validation block

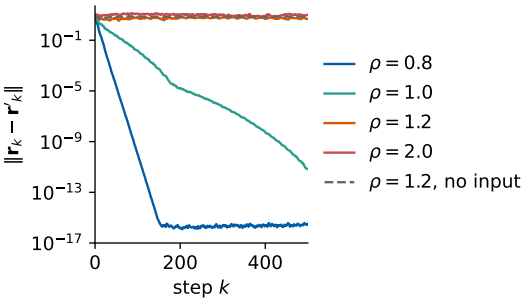
Effect of the Leak



the same four input pulses g_k (dotted) through three units; state x_k against the step k

- no leak, $x_k = x_{k-1} + \alpha g_k$: sums every past input, never forgets
- leak, $x_k = (1-\alpha)x_{k-1} + \alpha g_k$: an input j steps old counts as $(1-\alpha)^j$, old inputs fade
- no memory, $x_k = g_k$: copies the present input

Echo State Property



Definition

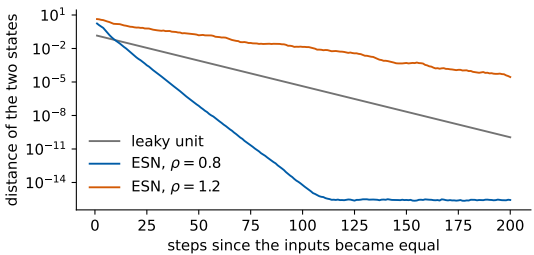
two copies, different starts $\mathbf{r}_0 \neq \mathbf{r}'_0$, same input s_k:

$$\|\mathbf{r}_k - \mathbf{r}'_k\| \to 0 \quad (k \to \infty)$$

- needed: a readout fitted on one run must work on another; also the reason for the washout
- sufficient ($\alpha = 1$): largest singular value of W below 1

figure: $N = 100$, Santa Fe input; scan: copies stay apart from $\rho = 1.00$ at $\sigma_{\text{in}} = 0.1$, from 1.45 at $\sigma_{\text{in}} = 2$ H. Jaeger, GMD Report 148 (2001)

Fading Memory



Definition

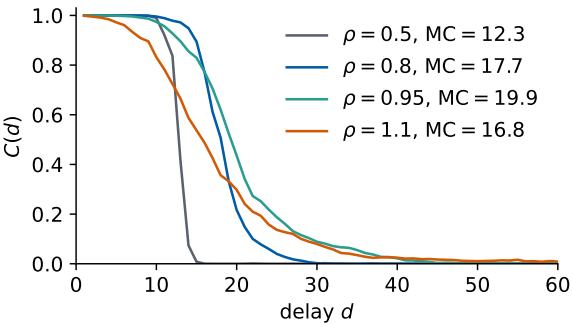
inputs $\mathbf{s}, \mathbf{s}'$ equal over the last K steps:

$$\|\mathbf{r}_k(\mathbf{s}) - \mathbf{r}_k(\mathbf{s}')\| \to 0 \quad (K \to \infty)$$

- needed: a fading-memory system is approximated by linear filters plus a polynomial readout (Boyd–Chua)
- echo state property: forget the start; fading memory: forget old inputs

figure: inputs differ for 50 steps, then coincide; median distance over 10 pairs; slope of ln distance: leaky unit $-0.105 = \ln(1-\alpha)$, ESN at $\rho = 0.8$ -0.333; ESN at $\rho = 1.2$: distance 2.8×10^{-5} after 200 steps S. Boyd, L. O. Chua, IEEE Trans. Circuits Syst. 32, 1150 (1985)

Memory Capacity



J. Dambre *et al.*, Sci. Rep. 2, 514 (2012)

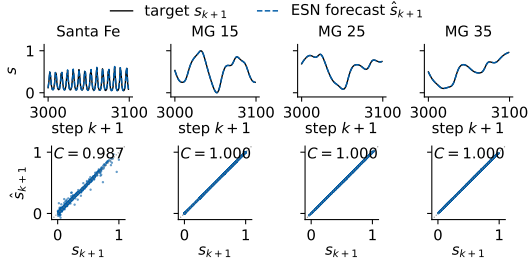
Recall of a random input

$$C(d) = r^2(\hat{u}_{k-d}, u_{k-d}), \quad \mathrm{MC} := \sum_{d \geq 1} C(d) \leq N$$

u_k: independent random input; one readout per delay d; MC: how many past inputs the state still holds

- one reservoir of 50 units: MC 12.3 at $\rho = 0.5$, 19.9 at $\rho = 0.95$
- MC measures memory of past inputs; C_{tot} measures forecasts

One-Step ESN Forecasts



one reservoir, $N = 300$; ρ, σ_{in}, α chosen on validation for each signal; top: target and forecast, bottom: $\hat{s}_{k+1}$ against s_{k+1}

- Santa Fe one step ahead: $C = 0.987$; over all 50 horizons the trained LSTM is better (C_{tot} 44.3 against 34.7)
- selected $\rho = 1.1$ for Lorenz and MG 15: the echo state property still holds for these inputs

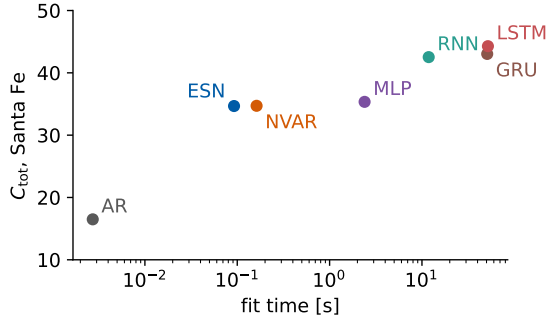
ESN against Trained Networks

model	trained	Lorenz	MG 15	MG 25	MG 35	Santa Fe
NVAR	readout	15.4	50.0	45.5	48.6	34.7
ESN	readout	39.2	50.0	48.3	49.8	34.7
MLP	all weights	29.4	50.0	48.4	49.5	35.3
RNN	all weights	43.8	50.0	49.4	49.5	42.5
LSTM	all weights	43.8	50.0	49.9	49.9	44.3

C_{tot} on the test block, maximum 50; same windows and split for all models

- Mackey–Glass: the ESN stays within 1.6 of the best trained network with only a readout fitted
- Lorenz and Santa Fe: the trained recurrent networks keep a clear lead

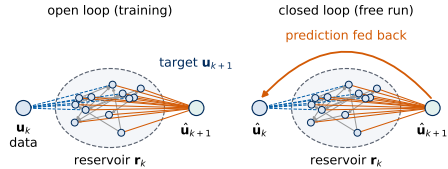
The Price of Training



test C_{tot} on the Santa Fe laser against the median fit time (logarithmic); one shared machine, times change with hardware and load

- ESN fit 92 ms, trained networks 2 to 52 s, 26 to 561 times longer
- trained networks keep a lead on the hardest records; the ESN costs a fraction

Closed-Loop Reservoir

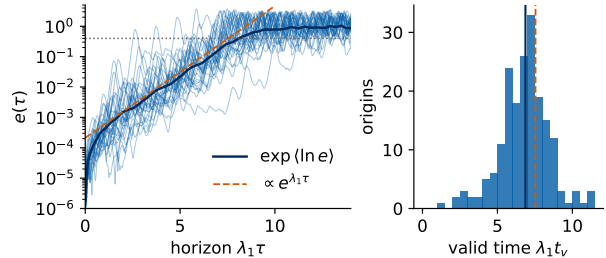


left: open loop, the measured state $\mathbf{u}_k$ drives the reservoir; right: closed loop, the forecast $\hat{\mathbf{u}}_{k+1}$ is fed back

- trained one step ahead on Lorenz-63 (x, y, z); closed loop: an autonomous system
- weather: how long the free run follows the true trajectory; climate: whether its long run has the statistics of the true attractor
- Lyapunov time $T_\lambda := 1/\lambda_1 = 1.11$; $\lambda_1 = 0.904$: rate at which nearby states separate

J. Pathak *et al.*, Chaos 27, 121102 (2017)

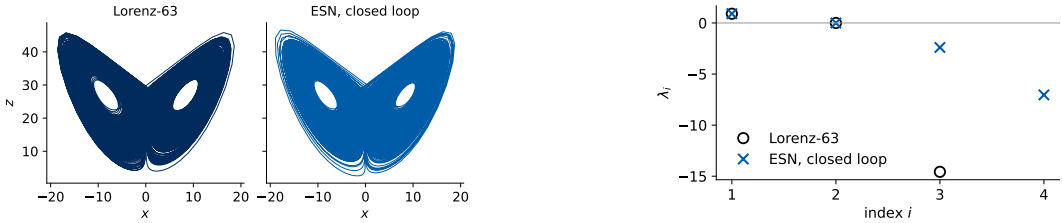
Valid Time of the Forecast



left: error $e(\tau)$ of the free run, distance to the truth over the attractor size ($e \approx \sqrt{2}$: unrelated points), against the horizon τ in units of T_λ, one line per origin, dashed: slope λ_1; right: histogram of the valid time t_v, first τ with $e > 0.4$

- median $t_v = 6.9\ T_\lambda$; 80 % of the 190 origins between 4.9 and 8.7 T_λ
- after a short initial jump the error grows at $0.93\,\lambda_1$, close to the rate of the Lorenz flow itself

Attractor and Lyapunov Exponents



left: (x, z) projection of a long true run and of the closed-loop ESN; right: Lyapunov exponents of the ESN against those of Lorenz-63

- densities of x, y, z: Wasserstein distance $W_1 = \int |P_p - P_q|\, ds$ (P_p, P_q: cumulative distributions) ≤ 0.016 standard deviations; two true runs differ by up to 0.015
- $\lambda_1^{\mathrm{ESN}} = 0.909$ against 0.904; $\lambda_2 \approx 0$ as well; λ_3 wrong: -2.40 against -14.57

Climate Depends on the Random Reservoir

	N=100, ρ=0.5	N=100, ρ=0.9	N=100, ρ=1.2	N=300, ρ=0.5	N=300, ρ=0.9	N=300, ρ=1.2
σ_{in}=0.1, β=10^{-8}	9	4	0	8	7	0
σ_{in}=0.1, β=10^{-4}	5	6	0	10	10	0
σ_{in}=0.5, β=10^{-8}	4	1	0	7	3	0
σ_{in}=0.5, β=10^{-4}	7	2	1	7	5	0

number of the 10 random reservoirs whose free run is Lorenz-like, for each setting of $N, \rho, \sigma_{\text{in}}, \beta$

- 96 of 240 runs Lorenz-like; the others: periodic (62), wrong chaos (44), fixed point (22), diverged (16)
- a small one-step error does not guarantee the right climate: check the free run

Requirements for a Good Reservoir

1. **echo state property**: the state forgets its start, depends only on the inputs
2. **fading memory**: recent inputs matter more than remote ones
3. **separability**: different recent histories give different states
4. **nonlinearity**: products of inputs at different delays
5. **richness**: many independent readable quantities
6. **time scales**: memory of the reservoir matches the task

- nothing here asks for a neural network: any driven system with these properties can serve

L. Grigoryeva, J.-P. Ortega, Neural Netw. 108, 495 (2018); G. Tanaka *et al.*, Neural Netw. 115, 100 (2019)

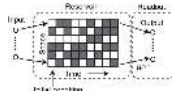
Physical Reservoirs

nothing inside is trained: any system with a nonlinear response, fading memory and many readable quantities, also interacting qubits; review: G. Tanaka *et al.*, Neural Netw. 115, 100 (2019)



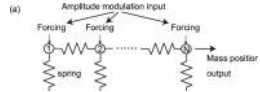
water in a bucket: the input drives waves, a camera image of the surface gives the features

C. Fernando, S. Sojakka, ECAL 2003, 588



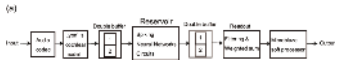
cellular automaton: input written into cells, a fixed local rule evolves them, the pattern is read out

O. Yilmaz, arXiv:1410.0162 (2014)



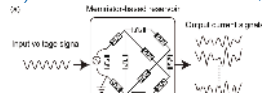
coupled oscillators: masses on springs driven by the input, positions read out

H. Hauser *et al.*, Biol. Cybern. 105, 355 (2011)



electronic circuit: one nonlinear node with a delay line, delayed samples act as virtual nodes

L. Appeltant *et al.*, Nat. Commun. 2, 468 (2011)



memristor network: resistance depends on the past current, so the network remembers

C. Du *et al.*, Nat. Commun. 8, 2204 (2017)

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Spiking Reservoir



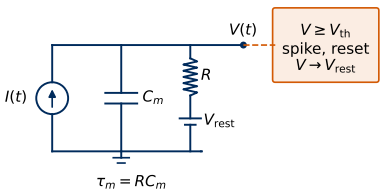
One Leaky Equation

$$\tau_r \frac{dx}{dt} = -x + \text{target}(t) \quad \xrightarrow{\text{Euler step } \Delta t} \quad x_k = (1-\alpha)\, x_{k-1} + \alpha\, \text{target}_k, \quad \alpha = \Delta t / \tau_r$$

	bucket, thin pipe	resistor–capacitor (RC) circuit	ESN unit i
state x	water level	voltage above rest v	r_i
storage	bucket	capacitor C_m	value of the last step
leak	outflow $\propto$ level	resistor R	factor $(1-\alpha)$ per step
target	inflow	$R\, I(t)$	$\tanh\left(\sum_j W_{ij} r_j + w_{\text{in},i}\, s_k\right)$
time constant	wide bucket, narrow pipe	$\tau_r = RC_m$	$\tau_r = \Delta t / \alpha$

- relaxes towards the target, forgets as e^{-t/τ_r}; the bucket is an analogy
- ESN: N such units wired by the random W; tanh limits each inflow
- neuron: the same leak, plus a spike and a reset at a threshold

Membrane as an RC Circuit



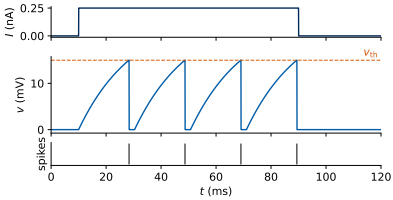
$$\tau_m \dot{V} = -(V - V_{\text{rest}}) + R\, I(t), \qquad \tau_m \coloneqq RC_m$$

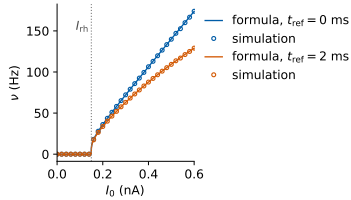
$$V(t) - V_{\text{rest}} = \frac{R}{\tau_m} \int_{-\infty}^{t} e^{-(t-t')/\tau_m}\, I(t')\, dt'$$

V: membrane voltage; C_m: capacitance; R: leak resistance; I: input current; $\tau_m = 20$ ms

- Kirchhoff: the input current charges the capacitor or leaks through the resistor
- the voltage is the input history weighted by $e^{-(t-t')/\tau_m}$: a fading memory

Leaky Integrate-and-Fire (LIF) Neuron





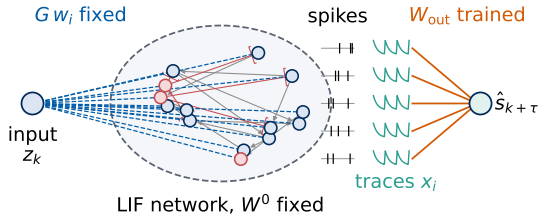
left: current step, voltage, spikes; right: rate against current, simulation (points), formula (lines)

$$\nu(I) = \left[t_{\mathrm{ref}} + \tau_m \ln \frac{RI}{RI - v_{\mathrm{th}}} \right]^{-1} \quad \text{for } I > I_{\mathrm{rh}} := v_{\mathrm{th}}/R, \quad v_{\mathrm{th}} = V_{\mathrm{th}} - V_{\mathrm{rest}}$$

- at V_{th}: spike, reset to rest, refractory pause t_{ref}; only spike times matter
- rheobase I_{rh}: smallest current that makes the neuron fire; formula and simulation agree within 0.6 %

L. Lapicque (1907), reviewed by L. F. Abbott, Brain Res. Bull. 50, 303 (1999)

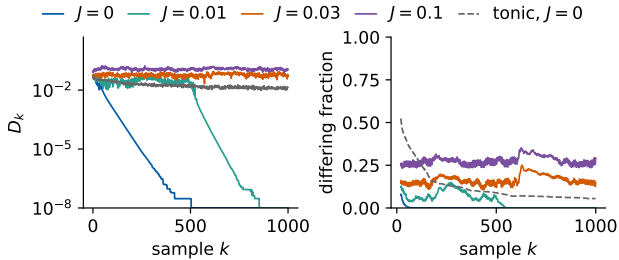
Liquid State Machine (LSM)



- 300 LIF neurons, fixed random synapses: a spike raises (excitatory) or lowers (inhibitory) the target voltage by J or $5J$
- input current $I_i = \theta_i + G\,w_i z_k$, $z_k = 2s_k - 1$, 10 ms per sample; θ_i: bias below threshold, G: gain
- features: spike trains smoothed into traces x_i, times $\tau_{s,i}$ from 5 ms to τ_s; ridge readout

W. Maass, T. Natschläger, H. Markram, Neural Comput. 14, 2531 (2002)

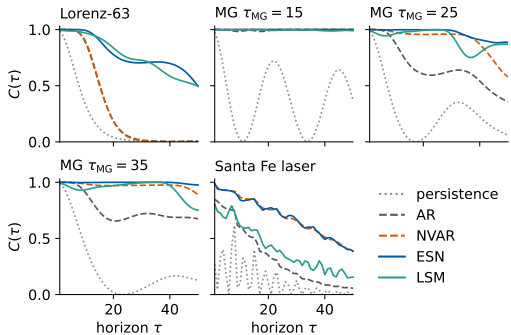
Echo State Property for Spikes



two copies, different starts, same input; left: relative distance D_k of the trace vectors, which decays after the spikes agree because the traces still carry earlier spikes; right: fraction of neurons whose spikes differ; J: coupling strength

- no coupling: identical spike trains after 59 samples; $J = 0.01$: after 532
- strong coupling ($J \geq 0.03$), or neurons that fire on their own (tonic): the copies never lock

Spiking Reservoir against the ESN

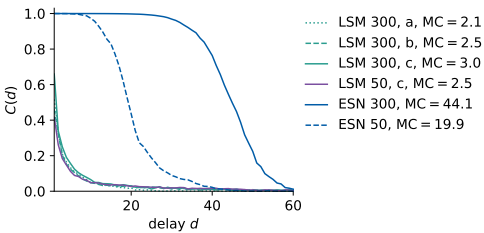


C_{tot}	AR	ESN	LSM
Lorenz	15.4	39.2	38.5
MG 15	49.8	50.0	49.7
MG 25	33.1	48.3	45.9
MG 35	38.0	49.8	47.2
Santa Fe	16.5	34.7	20.5

- smooth records: close to the ESN
- Santa Fe: far below; a fast pulse gets only a few spikes per sample

capacity $C(\tau)$ on the test block against the horizon τ; table: C_{tot}, maximum 50; ESN with 300 units, LSM with 300 neurons

Ingredients of the Spiking Reservoir



recall capacity $C(d)$ of a random input against the delay d; LSM a, b, c: three settings

- short memory: MC 2.1 to 3.0, ESN 44.1; $C(0) = 0.978$, $C(1) = 0.659$
- coupling does not help: $J = 0$ best or tied; a bank of uncoupled neuron filters
- spread of trace times essential: one common τ_s drops MG 25 from 45.9 to 20.3

one design, one encoding, a small grid; five random networks within 2.2 in C_{tot}

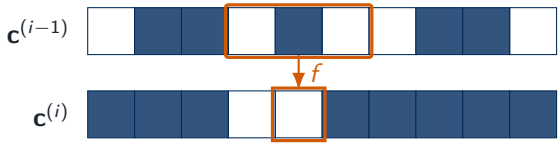
4

Cellular Automaton Reservoir



JAGIELLONIAN UNIVERSITY
IN KRAKOW

Cellular Automaton (CA)



ring: cell $L-1$ next to cell 0

Definition

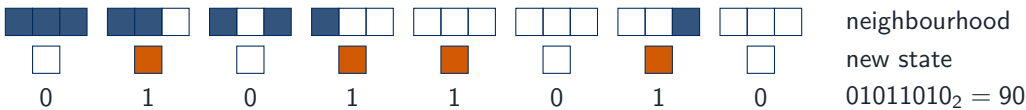
$c_j^{(i)} \in \{0,1\}$: state of cell $j = 0, \ldots, L-1$ of a ring after i iterations

$c_j^{(i)} = f\left(c_{j-1}^{(i-1)}, c_j^{(i-1)}, c_{j+1}^{(i-1)}\right)$, all cells at once, the same local rule f

global map $\mathbf{c}^{(i)} = F\left(\mathbf{c}^{(i-1)}\right)$: discrete in space, time and states

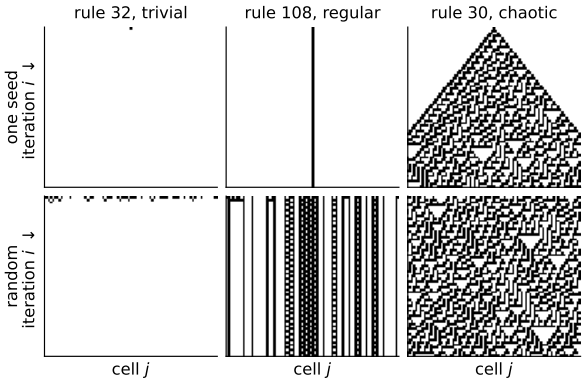
S. Wolfram, Rev. Mod. Phys. 55, 601 (1983)

Elementary Rules and Rule Numbers



- $q = 2$ states, radius $a = 1$ (three cells): $2^3 = 8$ neighbourhoods, $2^8 = 256$ elementary rules
- the 8 output bits, read as a binary number, give the rule number
- rule 90: $c_j^{(i)} = c_{j-1}^{(i-1)} \oplus c_{j+1}^{(i-1)}$, exclusive or (XOR), addition mod 2: a linear rule

Trivial, Regular, Chaotic



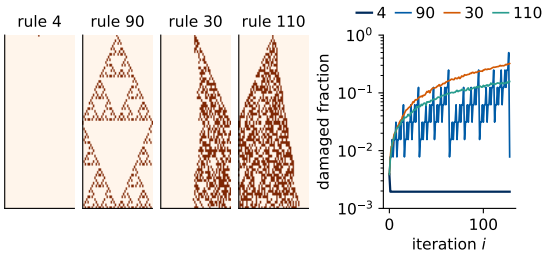
ring of 81 cells, 60 iterations i downwards, black: state 1; top: one seed, bottom: random start

- rule 32: empty from iteration 3
- rule 108: period 2 from iteration 2
- rule 30: no repeat within 60 iterations

Wolfram classes: I fixed (32); II stable or periodic (4, like 108); III chaotic (30, 90, 150); IV moving structures (110)

S. Wolfram, Physica D 10, 1 (1984)

Damage Spreading



Spreading rate

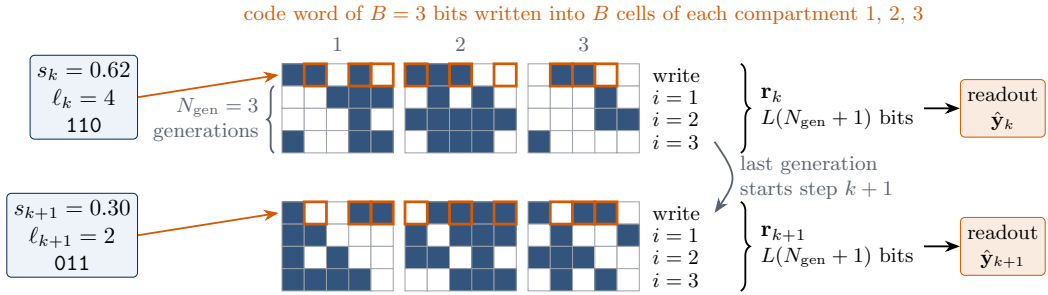
$$v_f := \frac{\text{width}}{2i+1}$$

width of the damaged region after i iterations, at most $2i+1$; $v_f \approx 0$: fixed width; loose analogue of λ_1

- rule 4: stays put; rule 90: nested triangles
- rule 30: right edge 1.00 cell per iteration, left edge 0.23

two copies differing in one cell; left: cells where they differ, i downwards; right: damaged fraction of 257 cells against i, mean of 20 starts

Cellular Automaton as a Reservoir

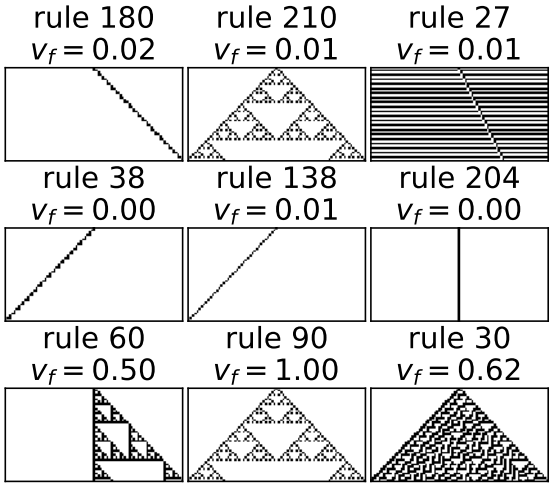


two input steps; rule 180, $Q = 8$ levels ℓ_k, Gray code, $R = 3$ compartments of 5 cells; orange: input cells, chosen once at random

- write the code word, iterate N_{gen} times; features $\mathbf{r}_k$: all rows; readout fitted
- memory: the other cells carry the past into the next step

O. Yilmaz, arXiv:1410.0162 (2014); J. Kantic *et al.*, Complex Intell. Syst. (2024)

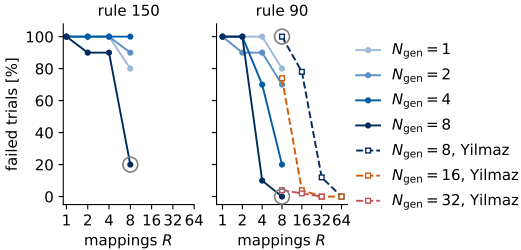
Three Fates of an Injected Bit



one bit on a quiet ring of 81 cells, 40 iterations downwards; above each panel: rule, spreading rate v_f on a random background

- travels (180, 38, 138, 27): a readable record of an earlier input
- spreads (90, 60, 30): mixes with later inputs; stays (204): overwritten by the next input
- $210 = c_{j-1} \oplus c_{j+1} \oplus c_j c_{j+1}$: rule-90 triangle from one seed, yet $v_f = 0.01$, since $c_j c_{j+1}$ acts on a random background

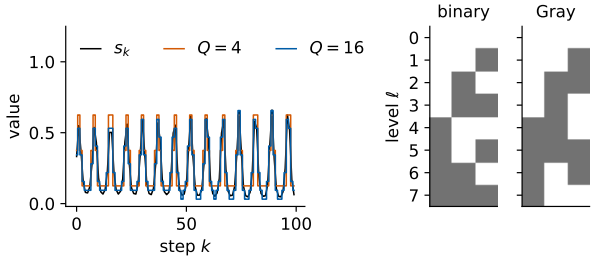
Five-Bit Memory Task



- 5 bits (32 patterns), then $T_0 = 200$ steps of distractor, then a cue
- then repeat the pattern; failed: any output off by more than 0.5
- at $R = N_{\text{gen}} = 8$: rule 90 fails in 0 %, rule 150 in 20 % of trials

failed trials (%) of 10 against the compartments R ("mappings"), one line per N_{gen}; right: rule 90, these runs (solid), Yilmaz (dashed); same trend, values differ: the paper leaves details open H. Jaeger, Jacobs Univ. Tech. Rep. 27 (2012); O. Yilmaz, arXiv:1410.0162

Encoding a Time Signal

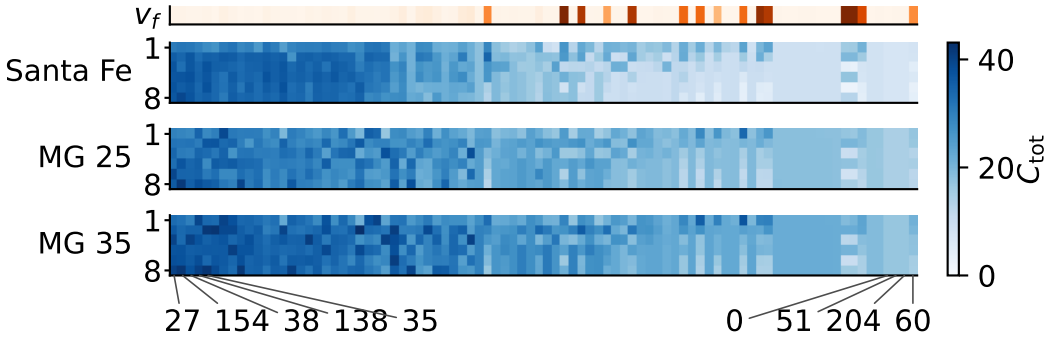


left: laser record s_k and quantised values $(\ell_k + 1/2)/Q$; right: code words of levels 0 to 7, grey for 1

- level $\ell_k = \mathrm{clip}(\lfloor Q s_k \rfloor, 0, Q-1)$, written as a code word of B bits, $Q = 2^B$
- Gray code: neighbouring levels differ in one bit (3→4: 011→100, Gray 010→110)
- rounding lost: root-mean-square error 0.0690, 0.0180, 0.0045 at $Q = 4, 16, 64$

J. Kantic *et al.*, Complex Intell. Syst. (2024), arXiv:2308.11522

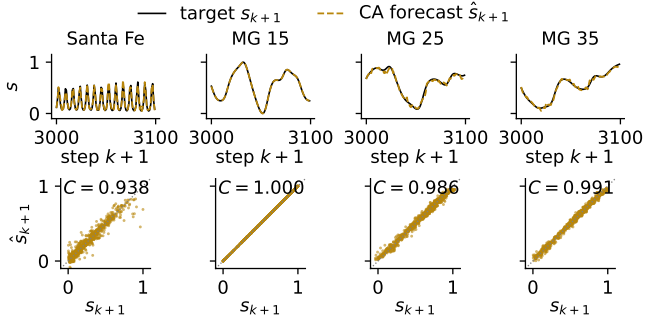
Forecast Quality of All Elementary Rules



validation C_{tot} of 88 rule classes (mirror, complement merged), best left, labels: five best, four worst; rows: $N_{\mathrm{gen}} = 1, 2, 3, 4, 6, 8$; strip: v_f, pale 0 to dark 1; $Q = R = 16$

- ten best: $v_f \leq 0.032$; class of 154 (with 166, 180, 210): top entry 39.1 (Santa Fe)
- last: 0, 51, 204 erase every earlier input; among slow rules v_f does not order quality

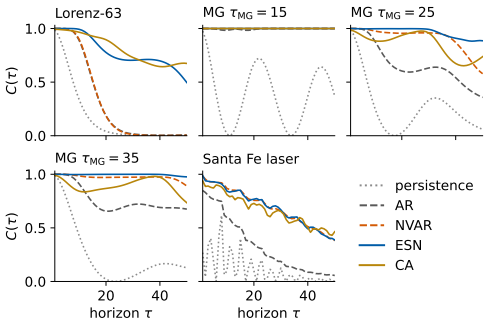
Cellular Automaton Forecasts



selected CA; top: target s_{k+1} and forecast $\hat{s}_{k+1}$; bottom: $\hat{s}_{k+1}$ against s_{k+1} on the test block, one-step capacity C

- one-step C: Santa Fe 0.938 (ESN 0.987), Mackey–Glass ≥ 0.986
- quantisation costs accuracy: the CA sees the laser only at 16 levels

Cellular Automaton against Echo State Network

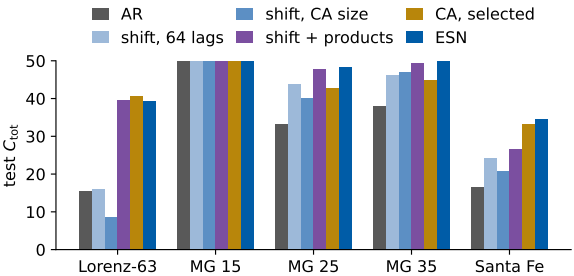


C_{tot}	ESN	CA, 320	CA, sel.
Lorenz	39.2	29.0	40.5
MG 15	50.0	49.9	50.0
MG 25	48.3	34.5	42.8
MG 35	49.8	40.9	44.8
Santa Fe	34.7	28.6	33.3

- 320 features: CA behind on every irregular record
- selected: above the ESN on Lorenz, close on Santa Fe, below on MG 25, 35

$C(\tau)$ on the test block against the horizon τ; table: C_{tot}; ESN: 300 features; selected CA: 768 to 2304

Shift-Register Control



- MG 25, 35: plain register 43.9, 46.3, CA 42.8, 44.8: transport of past values suffices
- Lorenz: register 16.0, with products 39.5, CA 40.5
- Santa Fe: register with products 26.5, CA 33.3

test C_{tot}; shift register: Gray-code bits of the last n_{lag} samples, zero code words before the record start, no automaton, same ridge readout; $n_{\mathrm{lag}} = 64$, as many features as the CA, or with products of bits of adjacent lags; one register design tested

Summary

reservoir	fixed state	finding
ESN	leaky tanh units, random W	near trained networks on Mackey–Glass, behind on Lorenz, Santa Fe; fit in ms
closed-loop ESN	the ESN fed its own forecast	median valid time 6.9 T_λ; right climate in 96 of 240 runs
spiking	leaky membranes with a threshold	short memory; a bank of single-neuron filters
cellular automaton	bits, a fixed local rule	best rules move bits without spreading

- common to all: a state that forgets its start and old inputs, only a linear readout trained

Thank you for your attention